

Education

Faculdade de Ciências e Tecnologia — Universidade Nova de Lisboa

Almada, Portugal

Integrated MSc in Computer Science — BSc: **14** (0-20); MSc: **15** (0-20)

December 2025

Thesis: **Formal Verification of Programs Equivalence** — **18** (0-20)

Experience

Opensoft

Lisbon, Portugal

Software Engineer

January 2026 – Present

- Migration of legacy applications to modern technologies (React, Kotlin, Java).
- Maintained multiple Spring and Java projects, resolving production and development issues.
- Increased the automatic test coverage of a project with 50k+ LoC by 5% using Junit and Mockito.

Fidelidade

Lisbon, Portugal

Software Engineer (Summer Intern)

July 2024 – August 2024

- Completed OutSystems training and resolved application issues.
- Design and initial development of a new version of an existing OutSystems application.
- Creation and presentation of a performance analysis report of several websites and webpages using Dynatrace.

Projects

Formal Verification of Programs Equivalence

Master's Thesis

- Developed bip2ml, a transpiler that transforms BipLang into OCaml, applying AST-level transformations to support equivalence-based program proofs.
- Built a local web app in HTML, CSS, JavaScript, and Python to visualize the input and output ASTs and support analysis of the transformation pipeline.

Campus Management Application

Bachelor's Final Project

- Worked in a team of five on a full-stack campus application using Java, Flutter, and Google Cloud Platform.
- Focused on back-end design, database definition, and server implementation, and the server was the most resilient out of 14 teams during load testing.

PicastloSocial

University Group Project

- Contributed to a layered, microservices-based social network built with Spring, Kotlin, H2, React, Redux, and TypeScript.
- Worked on a system with OpenAPI 3.0 documentation and full-stack integration between the back end and the front end.

Skills

- **Main Technologies:** Java, Kotlin, Spring Boot, TypeScript, JavaScript, React, Python, OCaml
- **Language:** English (C1), Portuguese (Native), Spanish (A2)

Certifications

- **Critical Thinking & Problem Solving** — Santander Open Academy — June 2025
- **Negotiation** — Santander Open Academy — June 2025
- **Escola das Finanças** — Magma Studio — March 2025
- **Public Speaking Skills Workshop** — Magma Studio — November 2025